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The Dark Side of Circuit Breakers

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1 Big picture

- **Question.** Do market-wide circuit breakers stabilize markets, or can the possibility of a trading halt itself destabilize prices as the market approaches the halt threshold?
- **Object of study.** The paper studies market-wide circuit breakers: rules that halt trading in the whole market when a benchmark index falls by a prespecified amount. In the current U.S. system, Level 1, Level 2, and Level 3 thresholds are tied to large percentage declines from the prior close.
- **Central mechanism.** A circuit breaker creates an endogenous illiquidity state. Once the halt is triggered, investors cannot rebalance until the market reopens or the terminal date arrives. This makes optimistic levered investors less willing to hold risky positions close to the trigger. The pessimistic investor therefore becomes more important in pricing the asset.
- **Core theoretical result.** In the model, the circuit breaker generally lowers the price level and sharply changes price dynamics near the trigger. As the market moves closer to the circuit breaker, conditional volatility rises, returns become more negatively skewed, expected returns rise, and trading activity increases.
- **Magnet effect.** The paper formalizes the “magnet effect”: near the threshold, the very existence of a circuit breaker raises the conditional probability that the market will hit the threshold. The channel is not investor psychology alone; it is an equilibrium volatility amplification channel.
- **Welfare result.** Circuit breakers are not unambiguously good or bad. If trading mainly reflects efficient risk sharing, a circuit breaker destroys market completeness and reduces welfare. If trading mainly reflects irrational speculation or panic, a circuit breaker can increase welfare by limiting harmful trades.
- **Empirical strategy.** Because market-wide circuit breakers are rarely triggered, the paper does not rely only on actual halt events. Instead, it tests model predictions using minute-level E-mini S&P 500 futures data from May 2013 through December 2020 and studies how volatility, skewness, returns, and volume behave as distance to the active circuit-breaker threshold shrinks.
- **Main empirical findings.** The data support the model’s predictions. Realized volatility rises as distance to the circuit breaker falls, and the slope becomes steeper close to the threshold. Skewness becomes more negative before reversing very near the boundary, one-minute expected returns rise, and abnormal volume increases.
- **Takeaway.** Circuit breakers may stop the market mechanically once the threshold is reached, but before the threshold they can make the market more fragile. The policy trade-off is therefore not simply “halts versus no halts,” but whether the benefits from stopping panic trading exceed the costs from distorted pre-halt trading and lost risk sharing.

2 Model, data, and measurement

2.1 Institutional object: market-wide circuit breakers

A market-wide circuit breaker temporarily closes trading in the broad equity market when a reference index falls sufficiently far from its prior close. The United States introduced market-wide circuit breakers after the 1987 crash. The U.S. rule was triggered in 1997 and then again during the COVID-19 market turmoil in March 2020. China introduced a market-wide circuit breaker in January 2016, but abandoned it quickly after it was triggered twice in the first week.

The policy motivation is stabilization: if prices are falling rapidly, a halt may give investors time to absorb information, reduce disorderly trading, and restore market order. The paper's point is that this logic misses an important equilibrium channel: the halt changes traders' incentives *before* the market closes.

2.2 Financial environment

The model is a continuous-time endowment economy on a finite interval $[0, T]$. There is one risky aggregate stock and one riskless bond. The stock pays terminal dividend D_T . The dividend process is

$$dD_t = \mu D_t dt + \sigma D_t dZ_t, \quad D_0 = 1, \quad (1)$$

where Z_t is a standard Brownian motion and $\sigma > 0$. The riskless bond is the numeraire and has terminal payoff one.

There is no intermediate consumption. The paper is therefore focused on intraday trading and rebalancing rather than long-run consumption smoothing.

2.3 Agents and beliefs

There are two agents, A and B . Agent A initially owns ω shares of the stock; agent B owns $1 - \omega$ shares. Let θ_t^i and ϕ_t^i denote agent i 's stock and bond holdings. Wealth is

$$W_t^i = \phi_t^i + \theta_t^i S_t, \quad dW_t^i = \theta_t^i dS_t. \quad (2)$$

Both agents have log utility over terminal wealth,

$$u_i(W_T^i) = \log(W_T^i), \quad i = A, B. \quad (3)$$

The two agents disagree about expected dividend growth. Agent A 's belief is the objective belief. Agent B 's expected growth rate is

$$\mu_t^B = \mu + \delta_t, \quad (4)$$

where disagreement follows

$$d\delta_t = -\kappa(\delta_t - \bar{\delta}) dt + \nu dZ_t. \quad (5)$$

When $\nu > 0$, agent B becomes more optimistic after positive dividend shocks and more pessimistic after negative shocks. In the important time-varying-disagreement case used in the simulations, large market declines make agent B more pessimistic near the circuit-breaker boundary.

The Radon-Nikodym derivative of agent B 's subjective probability measure relative to the objective

measure is

$$\eta_t = \exp \left(\frac{1}{\sigma} \int_0^t \delta_s dZ_s - \frac{1}{2\sigma^2} \int_0^t \delta_s^2 ds \right). \quad (6)$$

2.4 Alternative interpretation: state-dependent utility

The heterogeneous-belief model can be reinterpreted as a common-belief model with state-dependent utility. In particular,

$$\mathbb{E}^B \left[\log(W_T^B) \right] = \mathbb{E} \left[\eta_T \log(W_T^B) \right]. \quad (7)$$

This equivalence matters because the same equilibrium prices can have different welfare interpretations. If the heterogeneity reflects real state-dependent preferences, trades are risk-sharing trades. If it reflects distorted beliefs, trades may be irrational speculation.

2.5 Circuit-breaker rule

The circuit breaker closes the market when the stock price hits a downside threshold:

$$\tau = \inf \{ t \geq 0 : S_t \leq (1 - \alpha)S_0 \}, \quad \alpha \in [0, 1]. \quad (8)$$

For simplicity, the model assumes that after the circuit breaker is triggered, the market remains closed until T . This is a stylized version of real-world halts, but it captures the key feature: after the halt, investors lose the ability to rebalance continuously.

2.6 Empirical data

The empirical analysis uses transaction-level data for E-mini S&P 500 futures from the CME. The sample period is May 1, 2013 to December 31, 2020. The start date matches the implementation of the current U.S. market-wide circuit-breaker regime.

The E-mini is useful because it is highly liquid, closely tracks the S&P 500 index, and has price-limit rules during regular trading hours that match the market-wide circuit-breaker thresholds. The analysis uses regular trading hours from 9:30 a.m. to 4:00 p.m. ET, when the market-wide circuit breaker is active.

2.7 Minute-level variables

The paper works at the minute frequency. It first constructs second-level E-mini prices using volume-weighted average transaction prices. It then constructs:

- **One-minute returns:** log returns over one-minute intervals.
- **Realized volatility:** two-scale realized volatility to reduce microstructure noise.
- **Realized skewness:** the realized skewness measure used in high-frequency asset-pricing work.
- **Abnormal volume:** minute volume relative to recent average volume.

Abnormal volume is defined as

$$\text{Abnormal Volume}_t = \frac{\text{Volume in minute } t}{\text{Average volume in the previous 6.5 trading hours}} - 1. \quad (9)$$

2.8 Distance to circuit breaker

The key empirical state variable is distance to the active circuit-breaker threshold:

$$\text{DTCB}_t = \frac{P_t^{\min} - (1 - \alpha_t)P_t^{\text{close}}}{P_t^{\text{close}}}. \quad (10)$$

Here $P_t^{\min}$ is the minimum E-mini price during minute t , P_t^{close} is the prior-day end-of-day E-mini VWAP, and α_t is the nearest active threshold. A smaller DTCB_t means the market is closer to triggering a circuit breaker.

The paper drops observations after 3:25 p.m. to avoid noise from the rule change in which Level 1 and Level 2 halts no longer occur near the close.

2.9 Leverage-effect controls

Because lower prices are mechanically associated with higher volatility in many markets, the empirical specifications control for a leverage-effect proxy:

$$\text{Lev}_t = \frac{\text{Average E-mini price over the past 60 minutes}}{\text{Average E-mini price over the past 21 trading days}}. \quad (11)$$

The nonlinear leverage control is

$$\text{QLev}_t = (\text{Lev}_t - 1)^2. \quad (12)$$

These controls help separate the model's circuit-breaker channel from the standard leverage effect.

3 Models and empirical specifications

3.1 Equilibrium without a circuit breaker

Without a circuit breaker, the market is dynamically complete. The equilibrium allocation solves a social planner problem:

$$\max_{\widehat{W}_T^A, \widehat{W}_T^B} \mathbb{E}_0 \left[\lambda \log \widehat{W}_T^A + (1 - \lambda) \eta_T \log \widehat{W}_T^B \right] \quad \text{s.t.} \quad \widehat{W}_T^A + \widehat{W}_T^B = D_T. \quad (13)$$

The planner weight is tied to the initial wealth distribution, $\lambda = \omega$.

The no-circuit-breaker stock price can be written as a harmonic average of the two single-agent valuations:

$$\widehat{S}_t = \left(\frac{\widehat{\omega}_t^A}{\widehat{S}_t^A} + \frac{\widehat{\omega}_t^B}{\widehat{S}_t^B} \right)^{-1}, \quad (14)$$

where $\widehat{\omega}_t^i$ is agent i 's wealth share and $\widehat{S}_t^i$ is the valuation in an economy populated only by agent i .

Interpretation. In normal continuous trading, both agents can rebalance dynamically. The price reflects both the optimistic and pessimistic valuations, weighted by wealth. Neither agent's valuation mechanically pins down the price at a boundary.

3.2 Equilibrium at the circuit-breaker trigger

Once the circuit breaker is triggered, the market closes and agents must hold their positions until T . At the trigger time τ , each agent solves

$$V^i(W_\tau^i, \tau) = \max_{\theta_\tau^i, \phi_\tau^i} \mathbb{E}_\tau^i \left[\log(\theta_\tau^i D_T + \phi_\tau^i) \right] \quad \text{s.t.} \quad \theta_\tau^i S_\tau + \phi_\tau^i = W_\tau^i. \quad (15)$$

Because agents cannot rebalance after closure, the Inada condition rules out levered or short stock positions at the closing instant. With zero net bond supply, both agents hold their wealth in the stock at τ .

The trigger-time stock price is therefore

$$S_\tau = \min \left\{ \hat{S}_\tau^A, \hat{S}_\tau^B \right\}. \quad (16)$$

Interpretation. The pessimistic investor becomes the marginal investor at market closure. This is the key dark-side mechanism. A circuit breaker does not just halt trading; it changes who prices the asset near the halt.

3.3 Price before the circuit breaker

Before the trigger, the market remains dynamically complete, but the terminal object for valuation is no longer always D_T . If the circuit breaker is triggered, the terminal object for the pre-trigger trading interval is the closing price S_τ . The market-clearing price satisfies

$$S_t = \left(\omega_t^A \mathbb{E}_t[S_{\tau \wedge T}^{-1}] + \omega_t^B \mathbb{E}_t^B[S_{\tau \wedge T}^{-1}] \right)^{-1}. \quad (17)$$

Interpretation. Even before the market closes, investors price the possibility that the future marginal price will be set by the pessimist. This expectation pushes down the current price and makes the price more sensitive to negative shocks.

3.4 Main theoretical predictions

The model implies five main predictions.

1. **Lower price level.** For a given fundamental D_t , the stock price with a circuit breaker is below the stock price without a circuit breaker:

$$S_t < \hat{S}_t. \quad (18)$$

2. **Higher and accelerating volatility.** As S_t approaches the trigger, conditional volatility increases. The increase is stronger near the threshold.

3. **Negative skewness.** Since volatility rises when prices fall, return distributions become increasingly negatively skewed as the market approaches the circuit breaker. Very close to the boundary, the lower bound can mechanically reverse some of this skewness.
4. **Higher expected returns.** Near the trigger, the price drop is driven partly by changing marginal-investor influence rather than only by worse fundamentals. The model therefore predicts higher expected returns close to the threshold.
5. **Higher volume.** Optimistic investors want to cut stock exposure before being trapped in a trading halt. With trading frictions or execution delay, that rebalancing motive predicts higher trading volume as distance to the trigger falls.

3.5 Formalizing the magnet effect

Let

$$p_h(S_t) = \mathbb{P}(\tau \leq t + h \mid \mathcal{F}_t) \quad (19)$$

be the conditional probability that the market hits the circuit-breaker threshold within horizon h . Let $\hat{p}_h(S_t)$ be the analogous hitting probability without the circuit breaker. The magnet effect is captured by

$$\Delta p_h(S_t) = p_h(S_t) - \hat{p}_h(S_t) > 0 \quad (20)$$

and by the increase in this gap as S_t approaches the threshold.

The paper emphasizes that the magnet effect should be understood as a *conditional dynamic* phenomenon, not merely as a higher unconditional probability of ever reaching a low price.

3.6 Binned empirical specification

The first empirical specification is a binned scatter regression:

$$y_{d,m,t} = \sum_i \beta_i \mathbf{1}\{\text{DTCB}_t \in \text{Bin}_i\} + a_d + b_m + c\text{Lev}_t + d\text{QLev}_t + \varepsilon_{dmt}. \quad (21)$$

Here $y_{d,m,t}$ is realized volatility, skewness, abnormal volume, or next-minute return. The fixed effects a_d and b_m absorb day effects and minute-of-day seasonality. The bins divide distance to the threshold into intervals, with special fine bins near zero.

Interpretation. This specification is a flexible way to visualize whether the data follow the model-predicted shape: volatility and returns rising, skewness falling then reversing, and volume increasing as DTCB shrinks.

3.7 Piecewise linear empirical specification

The main statistical specification is

$$y_{d,m,t} = \beta \text{DTCB}_t + \gamma \min\{\text{DTCB}_t - 2\%, 0\} + a_d + b_m + c\text{Lev}_t + d\text{QLev}_t + \mathbf{Y}_t' \delta + \varepsilon_{dmt}. \quad (22)$$

For expected returns, the dependent variable is the next-minute return $r_{d,m,t+1}$. The vector $\mathbf{Y}_t$ contains lags of the dependent variable, with the lag length chosen by an AIC-based autoregressive

selection procedure.

Interpretation. The coefficient β captures the baseline relationship between distance to the circuit breaker and the outcome. The coefficient γ captures whether the relationship changes when the market is within 2% of the threshold. Since smaller DTCB means closer to the threshold, a negative coefficient on DTCB for volatility means volatility rises as the market moves closer to the circuit breaker.

3.8 Welfare measurement

The paper measures welfare using ex ante certainty-equivalent wealth. For agent A ,

$$\log(\text{CE}_0^A) = \mathbb{E}_0[\log(W_T^A)]. \quad (23)$$

For agent B , the welfare measure depends on the interpretation.

Under state-dependent utility,

$$\log(\text{CE}_0^B) = \mathbb{E}_0[\eta_T \log(W_T^B)]. \quad (24)$$

Under irrational subjective beliefs evaluated by a paternalistic planner under the objective probability,

$$\log(\text{CE}_0^B) = \mathbb{E}_0[\log(W_T^B)]. \quad (25)$$

The welfare effect of the circuit breaker is

$$\Delta \text{CE}^i = \text{CE}_0^i - \widehat{\text{CE}}_0^i, \quad \Delta \text{CE} = \omega \Delta \text{CE}^A + (1 - \omega) \Delta \text{CE}^B. \quad (26)$$

Interpretation. The same price dynamics can have different welfare meanings. If heterogeneity reflects useful risk sharing, the circuit breaker is costly. If heterogeneity reflects harmful mistaken beliefs, the circuit breaker can be beneficial.

4 Identification, interpretation, and credibility

4.1 What is identified empirically?

The empirical part is not a randomized policy experiment. The paper does not claim that DTCB is randomly assigned. Instead, the empirical exercise is a model-guided test of distinctive dynamic predictions. The model predicts a particular set of shapes near the circuit-breaker boundary, and the paper asks whether high-frequency data display those shapes after controlling for obvious alternatives.

This is an important distinction. The empirical analysis supports the model's mechanism; it is not a standalone causal estimate of the total welfare effect of circuit breakers.

4.2 Why actual circuit-breaker events are not enough

Market-wide circuit breakers are rare by design. A design based only on actual halt days would have very few observations and would mix the circuit-breaker mechanism with unusual macroeconomic

events. The paper instead uses the model’s local predictions: even when the market does not hit the threshold, the model predicts how prices and volume should behave as the market gets closer to it. This approach turns a rare-event problem into a high-frequency state-dependence problem.

4.3 Main threat: standard leverage effect

A natural concern is that volatility rises when prices fall even without any circuit breaker. The paper addresses this concern by including Lev_t and $QLev_t$, which compare recent E-mini prices to a 21-trading-day moving average. The key evidence is not only that volatility rises when prices fall, but that the relation between volatility and DTCB becomes steeper close to the circuit-breaker threshold.

The nonlinear steepening near 2% is the distinctive circuit-breaker prediction. A generic leverage effect would not automatically imply this kink around the policy boundary.

4.4 Fixed effects and lag controls

The regressions include day fixed effects and minute-of-day fixed effects. Day fixed effects absorb market-wide conditions on a given trading day, while minute fixed effects absorb intraday patterns such as the U-shaped pattern in volatility and volume.

The piecewise linear regressions also include lagged values of the dependent variable to address serial correlation in high-frequency outcomes. The paper reports both OLS and Newey-West standard errors.

4.5 Remaining limitations

- **Extreme observations are scarce.** The number of observations within 2% of the circuit breaker is small relative to the full sample and clustered in a few crisis windows.
- **The model abstracts from order-book liquidity.** The empirical figure shows a final dip in abnormal volume very close to the boundary, which may reflect deteriorating liquidity, not modeled trading demand.
- **The welfare analysis is qualitative.** The paper is explicit that the model is not calibrated to deliver definitive policy thresholds. The welfare trade-off depends on the relative importance of risk sharing, irrational speculation, and other market frictions.
- **Circuit breakers are only one intervention.** The same logic can apply to price limits, trading freezes, or other conditional restrictions, but different mechanisms and institutional designs may change the magnitudes.

4.6 Positive bond supply robustness

The baseline model assumes zero net bond supply. Section VI relaxes this assumption. With positive bond supply, the trigger-time marginal investor need not always be the pessimist. If the optimistic agent is rich enough to hold the whole stock market without leverage, the optimist can become marginal at the halt.

This extension shows that the volatility-amplification result is not mechanically universal. Still, for a wide range of bond-supply and wealth-share parameters, especially when net riskless bond supply is small relative to stock supply, circuit breakers continue to amplify volatility.

5 Tables and figures

5.1 Table I: Market-wide circuit breaker adoption

Read. Table I lists leading developed and developing equity markets and indicates whether they use market-wide circuit breakers and individual-stock price limits. The United States, United Kingdom, France, Canada, India, and China have used market-wide circuit breakers; China is marked as adoption followed by abandonment. Japan and Hong Kong are shown without market-wide circuit breakers in the table.

Economic message. Circuit breakers are not a niche U.S. institution. They are a major component of global market architecture. This motivates a general equilibrium analysis rather than a case-specific institutional anecdote.

Table I
Market-Wide Circuit Breaker Adoption

For the leading stock markets by market capitalization in 2020 (source: World Bank), the table indicates those markets that have implemented market-wide circuit breakers (MWCBS) and those markets that have implemented individual stock price limits. Y indicates adoption, N indicates no adoption, and Y/N for China denotes its adoption and then abandonment of a circuit breaker.

	Market Cap (Tn \$)	Market Cap Rank	Circuit Breaker	Price Limit
<i>Developed markets</i>				
United States	40.7	1	Y	Y
Japan	6.7	3	N	Y
Hong Kong	6.1	4	N	N
United Kingdom	3.6	5	Y	Y
France	2.8	6	Y	N
Canada	2.6	7	Y	N
<i>Developing markets</i>				
China (mainland)	12.2	2	Y/N	Y
India	2.6	8	Y	Y

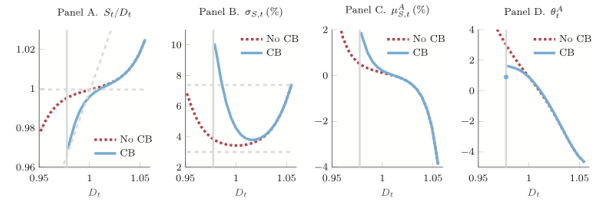


Figure 1. Price-dividend ratio, return volatility, instantaneous expected return, and agent A's portfolio. In the case of time-varying disagreement case, we plot all quantities as functions of dividend (D_t) at $t = 0.1$. In each panel, the gray vertical lines denote the circuit breaker threshold D_t , and the gray dashed lines correspond to the two cases with only one of the agents present in the market. The variables of interest are conditional on D_t . The parameters are $T = 1$, $\mu = 0.04\%$, $\sigma = \nu = 3\%$, $\kappa = 0$, $\delta_0 = \delta = 0$, $\omega = 0.9$, and $\alpha = 5\%$. (Color figure can be viewed at wileyonlinelibrary.com)

5.2 Figure 1: Model-implied price-dividend ratio, volatility, expected return, and holdings

Read. Figure 1 plots model-implied quantities as fundamentals decline under time-varying disagreement. As the market approaches the circuit breaker, the price-dividend ratio falls, conditional volatility rises sharply, the objective expected return rises, and the optimistic agent reduces stock exposure.

Economic message. The circuit breaker affects the whole pre-halt equilibrium. The critical force is not merely the mechanical halt; it is the anticipation of illiquidity and the shift in marginal pricing power toward the pessimistic investor.

5.3 Figure 2: The magnet effect

Read. Figure 2 compares the conditional probability of reaching the threshold with and without the circuit breaker. The probability with the circuit breaker rises faster and becomes much larger as the price approaches the threshold.

Economic message. The magnet effect is an equilibrium consequence of volatility amplification. Near the boundary, the circuit breaker makes the boundary more likely to be hit.

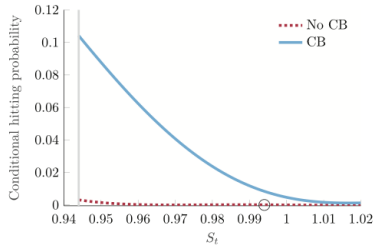


Figure 2. The “magnet effect.” In the case of time-varying disagreement, we plot the conditional probabilities for the stock price to reach the circuit breaker threshold within the next h minutes from $t = 0.1$. Here, h is proportional to $(S_t - S)^2$: h is roughly 0.5 minutes when $S_t = 0.95$, 9.5 minutes when $S_t = 0.97$, and 62 minutes when $S_t = 1.01$. The black circle represents S_0 . The gray vertical bar denotes the circuit breaker threshold S . The parameters are $\mu = 0.04\%$, $\sigma = \nu = 3\%$, $\kappa = 0$, $\delta_0 = \delta = 0$, $\omega = 0.9$, and $\alpha = 5\%$. (Color figure can be viewed at wileyonlinelibrary.com)

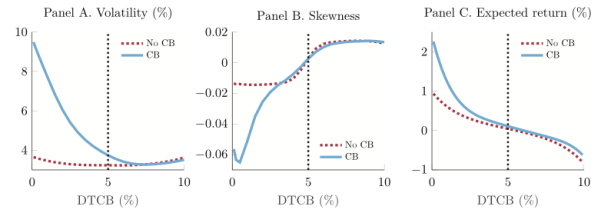


Figure 3. Model predicted one-minute conditional volatility, conditional skewness, and conditional expected return. In the case of time-varying disagreement, we plot all model predicted quantities as functions of the distance to the circuit breaker threshold (DTCB, defined as $S_t/S_0 - (1 - \alpha)$). Both volatility and return are normalized to one day. The parameters are $\mu = 0.04\%$, $\sigma = \nu = 3\%$, $\kappa = 0$, $\delta_0 = \delta = 0$, $\omega = 0.9$, and $\alpha = 5\%$. (Color figure can be viewed at wileyonlinelibrary.com)

5.4 Figure 3: Model-predicted one-minute return moments

Read. Figure 3 plots model-predicted one-minute volatility, skewness, and expected return against DTCB. Volatility rises dramatically and at an accelerating rate as DTCB falls. Skewness becomes more negative before reversing very close to the boundary. Expected returns rise close to the threshold.

Economic message. Figure 3 translates the model into directly testable empirical predictions. The paper can test the mechanism even without many actual circuit-breaker triggers.

5.5 Figure 4: Welfare impact

Read. Figure 4 shows welfare changes from introducing a circuit breaker under two interpretations. Under risk sharing, both agents and the market lose welfare. Under irrational speculation evaluated by the objective probability measure, the irrational agent gains from restricted trading, and total welfare can increase.

Economic message. Circuit breakers are welfare-reducing when they block beneficial risk-sharing trades, but welfare-improving when they block harmful speculative trades. The welfare answer depends on why investors trade.

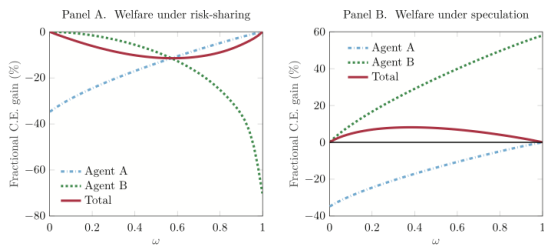


Figure 4. Welfare impact. Agents' gain in certainty equivalent wealth in a market with the circuit breaker versus a market without for different initial wealth shares ω of agent A (the case of time-varying disagreement). The blue dashed line corresponds to agent A and the green dotted line corresponds to agent B. The red solid line corresponds to the entire market, with proper weights for the agents. In Panel A, the certainty equivalent wealth is defined under agents' own beliefs. In Panel B, it is defined under the objective probability measure. The parameters are $\mu = 0.04\%$, $\sigma = 3\%$, $\nu = 6\%$, $\kappa = 0$, $\delta_0 = \delta = 0$, $\omega = 0.9$, and $\alpha = 5\%$. (Color figure can be viewed at wileyonlinelibrary.com)

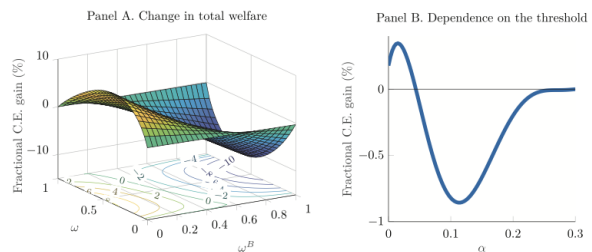


Figure 5. Welfare trade-offs. In the case of time-varying disagreement, the left panel shows the change in total welfare due to the circuit breaker in the presence of two types of trading motives: risk-sharing and irrational speculation. The right panel shows the dependence of the change in total welfare on the circuit breaker level. The parameters are $\mu = 0.04\%$, $\sigma = 3\%$, $\nu = 6\%$, $\kappa = 0$, and $\delta_0 = \delta = 0$. In the left panel, $\alpha = 5\%$; in the right panel, $\omega = 0.50$ and $\omega^B = 0.41$. (Color figure can be viewed at wileyonlinelibrary.com)

5.6 Figure 5: Welfare trade-offs and circuit-breaker design

Read. Figure 5 extends the welfare analysis to three agents: a rational objective agent, a rational state-dependent-utility agent, and an irrational-belief agent. The net welfare effect varies with the wealth shares and with the circuit-breaker threshold α . In the example, welfare is maximized at an interior threshold around 1.2%, but this is illustrative rather than a policy prescription.

Economic message. The policy problem is a trade-off. A tighter halt can reduce harmful speculation but also impede valuable risk sharing. There is no universal optimal threshold in the model.

5.7 Figure 6: Empirical binned scatterplots

Read. Figure 6 plots coefficients from the binned regression. Volatility rises as DTCB declines and the relation steepens close to zero. Expected returns also rise as DTCB declines. Skewness first falls as the market approaches the threshold and then reverses near the boundary. Abnormal volume rises as DTCB declines, although it dips in the final bin.

Economic message. The high-frequency data broadly match the model's qualitative predictions. The final volume dip likely reflects liquidity deterioration very close to the threshold, a feature outside the baseline model.

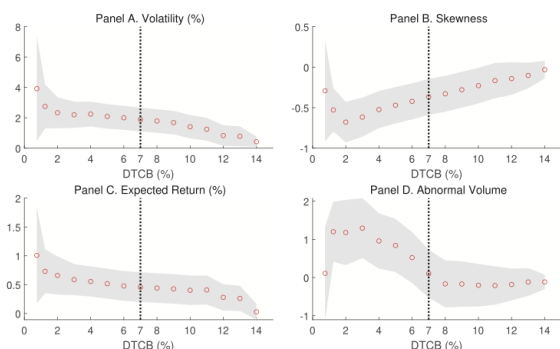


Figure 6. Binned scatterplot on regression coefficients of (40). This figure shows the β regression coefficients of (40) for volatility (two-scale realized volatility), skewness, expected return, and abnormal volume. $\text{Bin}_1 = (0\%, 0.75\%)$, $\text{Bin}_2 = (0.75\%, 1.25\%)$, $\text{Bin}_3 = (1.25\%, 2\%)$, and $\text{Bin}_i = ((i-1)\%, i\%)$ for $i = 4, \dots, 14$. The red dots indicate β coefficients and the gray regions represent two standard deviations around the β coefficients. (Color figure can be viewed at wileyonlinelibrary.com)

Table II
Regression results

This table reports results of the regression $y_{d,m,t} = \beta \times \text{DTCB}_t + \gamma \times \min\{\text{DTCB}_t - 2\%, 0\} + a_d + b_m + c \text{Lev}_t + d \text{QLLev}_t + Y_t + \varepsilon_{dmt}$, where $y_{d,m,t}$ is the realized volatility, skewness, and abnormal volume for day d , minute m , and time t or the return for time $t+1$. DTCB_{*t*} is the distance to the circuit breaker for time t , Lev_t is the leverage variable measuring the ratio between a short-term and long-term average E-mini prices at time t , $\text{QLLev}_t = \text{Lev}_t - 1\%$, and Y_t is a vector of lagged values of $y_{d,m,t}$. OLS and Newey-West standard errors (with 29 lags) are reported in parentheses and square brackets, respectively. Sample period: May 1, 2013 to December 31, 2020. The total number of observations is 686,697. $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Volatility			Skewness			Return			Abnormal volume		
Lev	-2.843 (0.176)***	8.555 (0.226)***	8.441 (0.229)***	-8.401 (0.237)***	-16.761 (0.238)***	-17.149 (0.231)***	-2.646 (0.376)***	1.116 (0.190)***	1.039 (0.197)***	0.166 (0.449)	24.004 (0.664)***	24.687 (0.669)***
QLLev	0.364*** (0.088)***	0.744*** (0.144)***	0.329*** (0.072)***	0.442*** (0.144)***	0.381*** (0.144)***	0.381*** (0.144)***	0.381*** (0.144)***	0.381*** (0.144)***	0.381*** (0.144)***	0.381*** (0.144)***	1.456*** (0.370)***	1.392*** (0.370)***
DTCB	-6.438 (1.289)***	2.075 (0.663)	2.590 (0.594)	-17.945 (0.894)	-23.863 (0.894)	-22.293 (0.894)	-1.911 (0.894)	1.104 (0.894)	1.406 (0.894)	8.391 (0.894)	26.328 (0.894)	24.074 (0.894)
$\min\{\text{DTCB} - 2\%, 0\}$	-17.454 (1.256)***	-17.263 (1.050)***	-12.958 (0.696)***	12.701 (0.696)***	5.375 (0.696)***	-5.254 (0.696)***	-28.299 (0.696)***	-29.277 (0.696)***	-1.656 (0.696)***	-1.543 (0.696)***	17.508*** (0.351)***	17.508*** (0.351)***
r^2	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Control for lags	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Date = Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Adjusted R^2	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001

5.8 Table II: Piecewise linear regression results

Read. Table II reports the regression

$$y_{d,m,t} = \beta \text{DTCB}_t + \gamma \min\{\text{DTCB}_t - 2\%, 0\} + \text{controls} + \varepsilon_{dmt}.$$

For volatility, the DTCB coefficient is about -17.5 , meaning that a 1% move closer to the circuit breaker raises daily-normalized realized volatility by about 17.5 bps. When $\text{DTCB} < 2\%$, the additional slope roughly doubles the effect in OLS estimates. For skewness, the baseline DTCB coefficient is positive, so skewness falls as the market moves closer to the threshold. For next-minute returns, the DTCB coefficient is negative, so expected returns rise near the threshold. For abnormal volume, the DTCB coefficient is negative, so trading activity rises as the threshold gets closer.

Economic message. Table II gives statistical support for the Figure 6 patterns. The evidence is strongest for the baseline relationships; inference on the near-threshold kink is more delicate because observations very close to the threshold are rare and autocorrelated.

5.9 Figure 7: Stock price with positive bond supply

Read. Figure 7 compares trigger-time stock prices with and without the circuit breaker when the riskless bond is in positive net supply. When the optimistic investor’s wealth share is not too high, the pessimist remains marginal and the circuit-breaker price is low. When the optimist is sufficiently wealthy, she can hold the stock market without leverage and can become marginal.

Economic message. Positive bond supply can soften or even reverse the baseline pricing effect in some parameter regions. The core mechanism depends on who becomes marginal when trading halts.

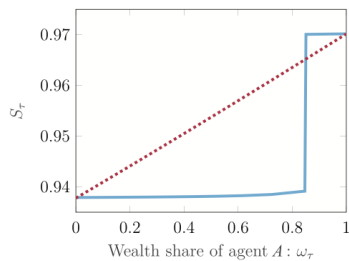


Figure 7. Stock price at time τ with positive bond supply. Blue solid lines correspond to the case with the circuit breaker. Red dotted lines correspond to the case without the circuit breaker. The parameters are $\tau = 0.25$, $D_\tau = 0.97$, and $\Delta = 0.17$. (Color figure can be viewed at wileyonlinelibrary.com)

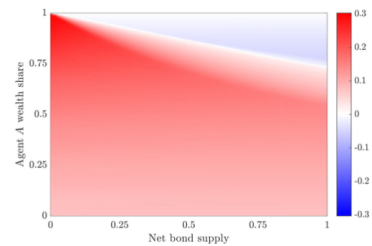


Figure 8. Impact of net bond supply on volatility. Heat map for the ratio of average daily volatilities in the economies with and without circuit breakers. For normalization, we subtract one from the ratios.

5.10 Figure 8: Positive bond supply and volatility amplification

Read. Figure 8 presents a heat map of average daily volatility ratios with and without circuit breakers across bond-supply and wealth-share parameters. Red regions indicate that the circuit breaker amplifies volatility; blue regions indicate the opposite.

Economic message. The volatility-amplification result is robust across broad parameter regions, especially when net bond supply is small. But the robustness section clarifies the economic condition behind the result rather than treating it as automatic.

6 Short summary

The paper builds a dynamic general equilibrium model of market-wide circuit breakers. Two agents trade because they disagree or because their utilities differ across states. Without a circuit breaker, the market is dynamically complete and prices aggregate both agents’ valuations. With a circuit breaker, the market becomes incomplete after the halt, and the pessimistic investor tends to become marginal at the trigger.

This shift in marginal pricing power feeds back into pre-trigger prices. As the market approaches the threshold, prices fall faster, volatility rises, returns become more negatively skewed, expected returns increase, and trading volume rises. This produces a formal magnet effect: the conditional probability of reaching the circuit breaker rises because the circuit breaker itself amplifies volatility

near the boundary.

The empirical analysis uses E-mini S&P 500 futures data from 2013 to 2020 and studies minute-level distance to the active circuit-breaker threshold. The observed behavior of volatility, skewness, expected returns, and volume is broadly consistent with the model.

The welfare result is nuanced. Circuit breakers reduce welfare when they block valuable risk-sharing trades. They can increase welfare when they limit irrational speculation. Hence, the same mechanism that creates a dark side in market dynamics can have either negative or positive welfare implications depending on investors' trading motives.

7 Conclusion

Core conclusion. Circuit breakers can destabilize markets before they stabilize them. Near a downside halt threshold, the anticipation of market closure changes investors' rebalancing incentives and shifts pricing power toward pessimistic investors. This raises volatility and increases the likelihood of hitting the threshold.

Economic intuition. The dark side comes from endogenous illiquidity. A trading halt makes future rebalancing impossible. Optimistic or levered traders therefore retreat before the halt, while pessimistic traders become more influential. The market price must adjust before the halt, not only after it.

Policy implication. Historical price data from periods without circuit breakers may understate the probability of triggering a circuit breaker, because the circuit breaker itself changes price dynamics. The design of halt thresholds should account for this equilibrium feedback.

Final takeaway. Circuit breakers are not simply emergency brakes. They are contingent trading restrictions that reshape incentives, prices, volume, and welfare before the brake is pulled.